

VAN-AN HOANG

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EXPERIENCE

Pixta - Largest marketplace online site of stock media in Asia

Junior/Middle AI Research Engineer

Hanoi, Vietnam

Oct 2023 - Aug 2025

- **AutoReview Project**

- **Engineered** a unified MLOps data pipeline using ClearML to integrate disparate data sources (AWS S3, DynamoDB, MySQL, Google BigQuery), enabling streamlined data processing for model training.
- **Evaluated, benchmarked, and integrated** multiple Vision-Language Models (including MiniCPM-V 2.6, Qwen2-VL, Gemini 2.0, Gemini 2.5) to current review system for abstract image understanding, focusing on trademark recognition and text overlay detection.
- **Analyzed** model performance against latency and inference cost, providing key data for deployment decisions in a real-time production environment.
- **Developed** a novel image processing algorithm for PNG&JPG files that automated a critical review step, recall NG 85%, recall accept 92%
- **Saved** approximately \$1,500 per month in manual review costs.

- **AI Search Project (Semantic Search Engine for Pixtastock)**

- **Architected** and deployed a semantic search backend using **OpenSearch** and **Weaviate** with HNSW-PQ indexing, which became the core of the publicly launched "AI Search" feature on pixta.jp.
- **Achieved a 90% search result acceptance rate** from the internal labeling team, validating the effectiveness of vision language fusion approach.
- **Deployed** autoscaling, asynchronous inference endpoints on **AWS SageMaker**, creating a robust CI/CD pipeline for system's deep learning modules.
- **Integrated** multi-modal models (**RAM++**, **Tag2Text**, **Osprey**) to generate rich semantic descriptions by fusing lexical tags with pixel-level visual understanding.
- **Built** a multi-stage detection pipeline combining COCO-baseline models with **YOLO-World** for open-vocabulary object detection and facial attribute recognition.

- **Independent Research & Technical Writing**

- **Authored** a technical blog post on **InstaFlow**, a one-step diffusion model, explaining its architecture and implementation. Published on Viblo, Vietnam's largest tech-sharing platform.
- **Researched** state-of-the-art models in diffusion and state-space architectures (DPM-Solver, Mamba, DiffuSSM) and built a PoC with Flux Kontext for a new image generation project.

DSLAb - BKAI International Research Center

Research Assistant

Hanoi, Vietnam

Nov 2022 - July 2024

- Efficient training diffusion models with SNR re-weighting and novel resampling strategy
Project link

- Resampled timestep with novel and more adaptive distribution
- Employed reversed Signal to Noise Ratio in Loss function
- Detailed Empirical Results with an acceleration in convergence speed and better FID, IS

- Researching on Deep Generative Models: Autoregressive Models, Flow Models, Generative Adversarial Network, Diffusion Probabilistic Models

Viettel Aerospace Institute

AI Engineer Intern

Hanoi, Vietnam

May 2023 - Sep 2023

- Redesigned body-face joint model github.com/AibeeDetect/BFJDet with a new post hoc for landmark detection (training code in pytorch) and write the evaluation code for landmark detection (metrics used: Mean Square Error, Root Mean Square Error)
- Ranked 27/165 Viettel Programming Challenge - A nationwide competition of Viettel Group on algorithm programming.
- Researched on Neural Radiance Fields (NeRF) and its Application on View Synthesis

EDUCATION

University of Technology Sydney, Sydney, Australia

Master of Artificial Intelligence

Aug 2025 — Now

Achievement and award

UTS Academic Excellence International Scholarship

Hanoi University of Science and Technology, Hanoi, Vietnam

Bachelor degree in Data Science and Artificial Intelligence

Oct 2020 — Oct 2024

Overall GPA: 3.66

Achievement and award

Excellent Scholarship - Type A (top 5%) - GPA: 3.93

The Scholarship Program Sponsored by the Electricity of Vietnam (EVN)

Machine Learning Teaching Assistant

April 2024 — June 2024

PROJECTS

VisionCraft AI - 3-day project in Tech Youth Builder Challenge

Hanoi, Vietnam

Project link

Oct 2025

- Developed a Large Language Model (LLM)-powered system for deep review, compliance checks, and market analysis of stock media.
- Implemented batch processing functionality and auto-generation of SEO-optimized titles, descriptions, and keywords, significantly streamlining content preparation.
- Integrated Image Generation capabilities (Style Transfer, Text to Image, Image to Image, Image Inpainting) to enhance the media creation and auto-enhancement from LLMs evaluation pipeline.
- Built a Retrieval-Augmented Generation (RAG) chatbot by crawling and indexing content from major stock guidelines (e.g., Adobe Stock, Shutterstock) to provide immediate, context-aware contributor support.

Efficient Training Diffusion Models

Hanoi, Vietnam

Project link

March 2024 - July 2024

- Resampled timestep with novel and more adaptive distribution
- Employed reversed Signal to Noise Ratio in Loss function
- Detailed Empirical Results with an acceleration in convergence speed and better FID, IS

Analysis on different methods of rain removal task

Hanoi, Vietnam

Project link

Oct 2022 - Jan 2023

- Fine-tuned Gated Context Aggregation Network on the task of removing rain drop on lens when capturing landscape and give a benchmark test on SSIM and PSNR to compare with other models such as DDPM, GAN, Transweather

Document Summarization

Hanoi, Vietnam

Project link

Dec 2022 - Jan 2023

- Implement Sequence to Sequence (Seq2Seq) RNN with pytorch
- Fine-tuning model google mt5-small on processed cnn.dailymail dataset and obtain Rouge-1: 28.102, Rouge-2: 13.426, Rouge-L: 24.824 (comparable to GPT-2), outperform implemented Seq2Seq model.

SKILLS

- **English:** IELTS 7.5 Overall
- **Programming:** Python (NumPy, Pandas , Scikit-learn, TensorFlow, PyTorch, Pytorch Lightning, Matplotlib, Plotly), Java, C++, SQL, Git, ElasticSearch, OpenSearch, Generative AI, ComfyUI, Flux, Stable Diffusion

REFERENCES

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